

# Non-Contact Thermal Haptics for VR

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## ABSTRACT

In the realm of virtual reality (VR), haptic feedback plays a pivotal role in enhancing user immersion. However, the implementation of non-contact thermal haptics presents numerous challenges. In this study, we introduce a novel approach for creating thermal haptic feedback in a VR environment, combining an ultrasonic phased array with a heating circuit. Our design generates multiple beams that deliver heated air to the user, thus simulating the sensation of temperature change. This approach enables an immersive, responsive VR experience that dynamically adjusts the temperature of the surrounding air based on the VR content. The innovation resides in our method's ability to overcome traditional phased array limitations, such as grating lobes and high hardware costs, by leveraging the directionality of ultrasonic transducers within the array. This results in a more cost-effective, compact, and efficient thermal haptic feedback system. Our work thus offers a significant contribution to the field of VR, presenting a new way of enhancing user immersion through non-contact thermal haptics.

## CCS CONCEPTS

• Human-centered computing → Haptic devices.

## KEYWORDS

thermal haptic, non-contact, virtual reality, temperature

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## 1 INTRODUCTION

Thermal haptics refers to the sensory feedback system that enables users to perceive temperature changes, enhancing the overall immersive experience in Virtual Reality (VR). As an example application shown in Fig. 1, the perception of heat or cold in a VR



**Figure 1:** In a practical application of our non-contact thermal display, consider a user immersed in a VR game where they must dodge fireballs from an adversary. Should the user fail to evade a fireball, a beam of heated air generated by our system would target them, consequently providing a sensation of heat.

environment contributes to a fuller, more realistic simulation of the physical world, effectively bridging the gap between the virtual and the real. By introducing thermal haptics into VR, we can deepen the level of interaction and engagement, thereby expanding the potential applications of VR in areas such as education, gaming, and training simulations.

Implementing non-contact thermal haptics in VR presents several challenges. One of the primary challenges is to create a sensation of temperature change without physical contact. This involves not only generating varying temperatures in the surrounding air but also directing the heated or cooled air towards the user's body parts accurately. Additionally, the system should be responsive enough to adjust the temperature in real-time according to VR content. Therefore, the design and integration of such a system require intricate calibration between different components and a sophisticated understanding of flow dynamics and thermal perception.

In response to the challenges of implementing non-contact thermal haptics in VR, we propose a novel method involving the synergistic use of an ultrasonic array and a heating circuit. The heating circuit plays a pivotal role in adjusting the surrounding air's temperature based on the interactive content of the VR environment. It is designed with the capability to generate a spectrum of temperatures, thus providing a diverse range of thermal experiences.

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Complementing the heating circuit, we introduce an ultrasonic array. This technologically advanced component is capable of generating multiple beams, akin to invisible 'hands', that can direct the air of different temperatures precisely towards the VR user's body. The array is meticulously calibrated to ensure that the heated or cooled air reaches the desired areas, such as the user's hands or other body parts.

Together, the heating circuit and ultrasonic array form a sophisticated system that provides a non-contact method of delivering temperature changes to the user. This system is not only responsive and precise but also adjustable, paving the way for a richer and more immersive VR experience.

Our proposed system offers several advantages over traditional methods of providing thermal haptics. Firstly, it realizes a non-contact form of thermal haptic feedback, eliminating the need for the user to wear additional equipment, which can be restrictive and uncomfortable. This significantly enhances the user's freedom of movement and overall comfort during the VR experience.

Secondly, our system provides a high degree of precision in directing the temperature changes to the user's body parts. The ultrasonic array's capability to produce multiple beams allows for the accurate delivery of varying temperatures to specific areas, creating a more realistic and immersive thermal haptic experience.

Lastly, the responsiveness of our system contributes to its advantage. The heating circuit's ability to adjust temperatures in real-time based on VR content creates a dynamic and responsive environment, thus maintaining seamless VR experience.

In this paper, we contribute to the field of VR and haptics in several significant ways. Firstly, we propose an innovative solution that successfully provides non-contact thermal haptic feedback in VR, addressing a long-standing challenge in the field. Our method of combining an ultrasonic array with a heating circuit offers a novel way to generate and deliver varying temperatures accurately.

Secondly, we delve into the technical aspects of our proposed system, shedding light on the intricate calibration and workings of the heating circuit and ultrasonic array. This detailed exposition of our design process provides valuable insights for future research and development in thermal haptics.

Lastly, we explore potential applications of our system, demonstrating its versatility and potential to enhance user interaction in a wide range of VR contexts. This broadens the horizon for the application of thermal haptics, opening up new possibilities for immersive and interactive VR experiences.

The remainder of the paper is organized as follows: Section 2 reviews related works. In Section 3, our system design is detailed, with specific focus on the ultrasonic array and the heating circuit. Section 4 explores potential applications. Finally, Section 5 concludes the paper and discusses avenues for future work.

## 2 RELATED WORK

Temperature presentation for VR has been extensively studied. The existing methods can be divided into contact and non-contact methods.

In contact-based methods, the temperature presentation requires users to directly touch or contact with the thermal components and the skin feel the thermal changes via heat conduction. The most

common strategy is to use Peltier elements [5, 13, 19, 21]. Other contact-based methods include using fluid flow [3, 20, 24], resistive heating [11], gel packs [7] and thermal conductivity control [4]. Contact-based methods have fast responsiveness and are able to deliver thermal haptics with fine spatial resolution and wide-range temperature change. However, the contacting medias are inconvenient to equip and keeps the users from fully immersive VR experience.

Non-contact methods are achieved via convection or radiation without direct contact with the heat source. A widely used strategy is to use infrared lamps [6, 9, 12, 22]. Ichihashi et al. apply infrared rays and shutter mechanism to deliver heat sensation with fast responsiveness [10]. However, their heat source is bulky and the heat delivery is limited with fixed angle and distance. Air flow can also be used for temperature presentation. Nakajima et al. apply ultrasound phased arrays to form directional cool air flow for cooling sensation [16–18]. However, their works are limited to cool air or mist with uncontrollable temperature and their cold air sources rely on storage tanks. Other non-contact methods include humidity adjustment [8], odor-based illusion [2] and visuo-thermal interaction [14, 15, 23]. Although these non-contact methods do not require direct-contact devices and therefore achieves better immersion, they suffer from slow responsiveness and low spatial resolution.

## 3 SYSTEM DESIGN

Fig. 2 illustrates our non-contact thermal display system, which is comprised of two key components: a heating circuit and an ultrasonic transducer array. The resistors of the heating circuit are placed around the ultrasonic transducers, generating heat to warm the surrounding air continuously. In operation as a thermal display, our system uses a heat map that corresponds to the current VR content as its input. It then triggers the corresponding ultrasonic transducers to emit multiple beams of ultrasonic waves, which carry the heated air towards the user, thereby simulating the sensation of temperature change. The design details of these two components are elaborated further below.

### 3.1 Heating Circuit

The heating circuit [1] is common in many appliances and designed to heat up the surrounding air according to the temperature requirements of the VR content and the distance from the circuit to the VR user. We estimate the required heating temperature considering the distance-induced temperature drop during the air's transit from the heating circuit to the user through the ultrasonic array.

The heating circuit operates based on voltage division between a resistor and an negative temperature coefficient (NTC) thermistor, controlling a MOSFET or a transistor. When the temperature drops below a certain threshold, the voltage division between the resistor and the NTC thermistor rises, activating the heating circuit. As the temperature increases, the voltage division decreases, and the circuit shuts off.

For the circuit's implementation, we have chosen a MOSFET in an SOT-23 package, specifically the *LN2302BLT1G* model from LRC, over a transistor due to the larger conduction current it can

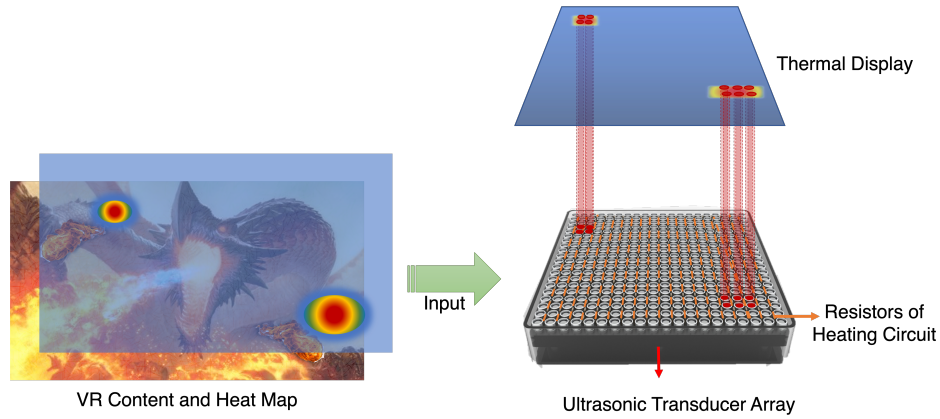


Figure 2: System prototype.

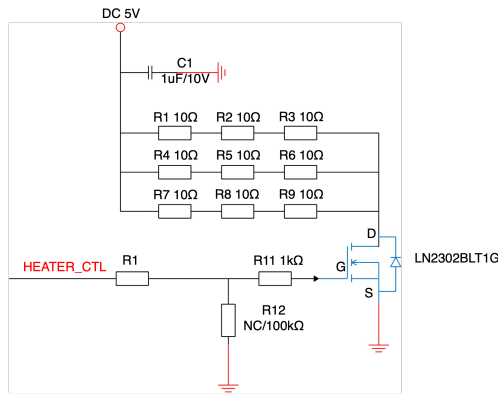


Figure 3: Heating circuit diagram.

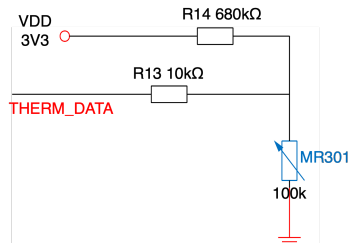


Figure 4: Temperature control circuit diagram.

handle in the same package. The NTC thermistor used is the commonly used *NCP15WF104F03RC*. The heater device is a 10-ohm resistor, keeping the total power of the heating circuit at 2.5W.

The signal *THERM\_DATA* serves as an analog signal to the ADC port of the microcontroller on the PCB, providing a measure of the PCB's temperature. *HEATER\_EN* is connected to the control IO of the microcontroller. If the heating circuit loses control, the IO is pulled high, forcibly shutting off the MOSFET as a safety measure.

The critical parameter in this circuit is the value of *R14*, which along with the voltage divider resistor *R13*, directly determines when the MOSFET *Q302* opens. We have set 40°C as the threshold for activating and deactivating the heating circuit. The resistor value at this temperature is 50.7K ohms, and the MOSFET activation voltage at this temperature is 0.9V. Using these values, we calculate the voltage divider resistor to be 680K.

The heating areas of the two main PCBs are in the same location, so we have concentrated the heating resistors in the corresponding

areas on the heating circuit sub-board. We have removed excess copper on the board, and the NTC resistor is placed on one of the main PCBs. We also added a thermal silicone pad between the heating area of the board and the two main PCBs to enhance heat conduction.

### 3.2 Ultrasonic Array

In our system design, we have chosen to use an ultrasonic array consisting of  $20 \times 20$  elements with transducers operating at a center frequency of 40kHz. We decided against employing a phased array for multiple beamforming due to several reasons.

- The size of each ultrasonic transducer in a phased array is larger than half a wavelength, which leads to the generation of grating lobes. These lobes cause unwanted temperature changes in areas where they should not occur. Although there are techniques to suppress grating lobes, they tend to weaken the main lobe's intensity or reduce the steering range, both leading to a smaller area for thermal haptics.
- To implement a  $20 \times 20$  phased array, each ultrasonic transducer requires a separate channel to control its phase, meaning we would need 400 phase shifters. This requirement significantly increases the hardware cost and the size of the device.

To overcome these challenges, we have taken a different approach. We leverage the inherent directionality of the ultrasonic transducers and treat the  $20 \times 20$  ultrasonic array as a  $20 \times 20$  thermal display. Each array element acts as a pixel of the display, regulating the temperature in its direction. Our preliminary studies show that due to the slow and continuous spatial changes in temperature caused by thermal diffusion, humans are less sensitive to changes in temperature and spatial variation. Therefore, we have set the distance between each element to be 3cm, enabling us to create a  $60\text{cm} \times 60\text{cm}$  thermal display within a 1m distance.

To address the issue of each array element requiring an independent channel, we designed an element control matrix as shown in Fig. 5. Each row of the ultrasonic array is connected to a switch grounded, and each column is connected to a switch linked to a sound source (i.e., an audio channel). Each of 40 switches (20 rows + 20 columns) is controlled by an IO port to turn on or off. To generate a beam from the ultrasonic transducer at the  $i$ -th row

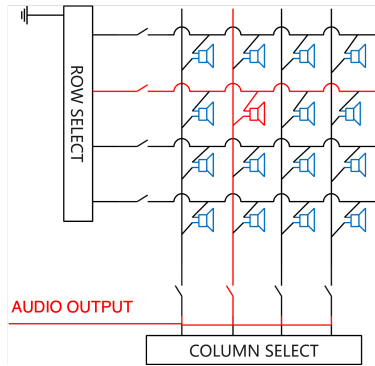


Figure 5: Element control matrix.

and  $j$ -th column and change the temperature of the corresponding thermal display pixel, we simply need to turn on the two corresponding switches in the  $i$ -th row and  $j$ -th column. When we need to change the temperature of multiple pixels, we use time-division multiplexing, rapidly switching between different arrays in units of 100ms.

Our proposed method has several advantages over phased array-based technology:

- **Simplified Hardware Configuration:** Our approach significantly reduces the hardware complexity and the number of required channels. This simplification results in lower hardware costs and smaller device size.
- **More Precise Temperature Control:** By treating each element of the array as a pixel in a thermal display, we can control temperature changes more precisely in the corresponding direction, reducing the occurrence of unwanted temperature variations.
- **Greater Scalability:** Our method is highly scalable, allowing for larger arrays without a substantial increase in complexity or cost.

## 4 APPLICATIONS

Thermal haptics can be effectively applied across a wide variety of fields, demonstrating its extensive versatility and utility.

**Virtual and Augmented Reality:** The advent of immersive technologies such as Virtual Reality (VR) and Augmented Reality (AR) has paved the way for multi-sensory experiences that extend beyond the traditional confines of visual and auditory stimuli. The proposed ultrasound array-based thermal display technology can significantly enhance the realism of these virtual environments by creating a rich sensory tapestry that includes precise spatial variations in temperature. This addition of a thermal dimension to VR and AR can facilitate a more immersive and emotionally engaging user experience, simulating environments ranging from a warm sunny beach to a chilly snowscape.

**Touchless Human-Machine Interaction:** In the realm of touchless human-machine interaction, the proposed technology can provide a novel means of feedback—thermal haptic feedback. The ability to manipulate spatial temperature variations can provide users with intuitive and immersive interactions, thereby elevating the overall user experience. This technology can serve as a bridge between users and digital interfaces, enabling a new form of non-contact, thermal haptic communication.

**Medical and Rehabilitation:** The proposed technology also has potential applications in the fields of healthcare and rehabilitation. By enabling the design of personalized thermal therapy programs, it could contribute to pain alleviation, muscle relaxation, and rehabilitation training. The precise control of thermal stimulation offers a non-invasive approach to address various health needs.

**Art and Entertainment:** In the art and entertainment industries, artists and designers can utilize this technology to create interactive pieces or exhibitions that incorporate temperature variations. This can add a new dimension to the audience's engagement and experience, stimulating not only their sight and hearing but also their sense of touch through temperature.

**Education:** The technology can be employed to create more interactive learning environments in educational settings. By simulating temperature variations, it could help students gain a more tangible understanding of concepts related to climate change, thermodynamics, or other natural science phenomena, making the learning experience more engaging and effective.

**Scientific Research:** The proposed system can also prove to be a valuable tool for scientific research. It can be used in studies related to temperature perception, human responses to temperature variations, and other related fields. This technology can offer researchers a means to explore and understand the complex interplay between thermal stimuli and human perception.

In summary, the proposed ultrasound array-based thermal display technology presents a wide array of application possibilities that extend across various domains. As this technology continues to evolve and mature, it is expected to open up new opportunities for creating more immersive, engaging, and intuitive experiences.

## 5 CONCLUSION AND FUTURE WORK

In this study, we have presented a novel approach to creating non-contact thermal haptic feedback in virtual reality (VR) environments. Our system, which combines an ultrasonic phased array with a heating circuit, demonstrates a unique method of delivering temperature changes to a VR user. By leveraging the intrinsic directionality of the ultrasonic transducers within the array, we have overcome traditional phased array limitations such as grating lobes and high hardware costs.

Our system not only enhances the immersive experience for VR users but also proves to be more cost-effective, compact, and efficient compared to traditional phased array technologies. This advancement in the field of non-contact thermal haptics marks a significant contribution to the world of VR, offering a new way of enhancing user immersion.

Looking to the future, several potential areas for further development and research can be identified. Optimizing the control matrix for larger arrays and refining the system's thermal control precision are among the technical improvements that can be addressed. Moreover, exploring additional applications, such as in medical training or simulation, could extend the reach of our technology. Lastly, investigating user perception and comfort levels with different temperature variations and durations would provide valuable insights for improving the overall user experience. Our research opens up a new path in the realm of VR and haptics, one which we are eager to continue exploring.

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